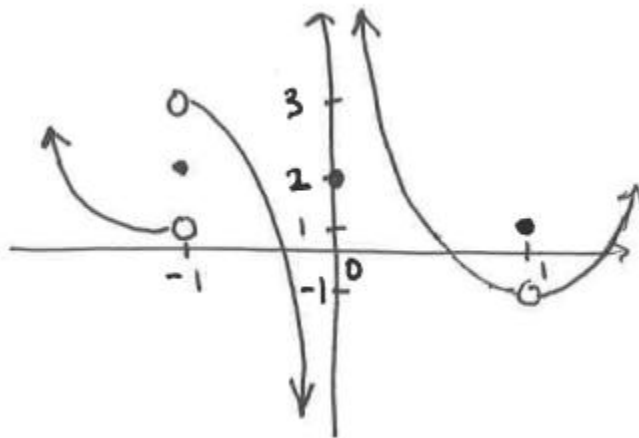


1. Consider the graph of the $f(x)$ shown below.



Determine each of the following.

- $\lim_{x \rightarrow -1^-} f(x)$
- $f(-1)$
- $\lim_{x \rightarrow -1} f(x)$
- $\lim_{x \rightarrow 0^+} f(x)$
- $\lim_{x \rightarrow 1} f(x)$
- $f(0)$

2. Create a function that has the following properties.

$$\lim_{x \rightarrow 0^+} f(x) = -\infty, \quad \lim_{x \rightarrow 0^-} f(x) = 4, \quad f(1) = 3, \quad \lim_{x \rightarrow 3} f(x) = 1, \quad f(3) = 2$$

3. Answer each of the following questions.

- a) Create a function that has the following properties.

$$\lim_{x \rightarrow 1} f(x) = 2, \quad f(1) \text{ is undefined}, \quad \lim_{x \rightarrow 3^+} f(x) = \infty, \quad \lim_{x \rightarrow 3^-} f(x) = -\infty, \quad f(-2) = -1, \quad \lim_{x \rightarrow -2^-} f(x) = 1$$

- b) Can you draw another function that meets all of the given conditions but has different behavior at the x -values of -2, 1, & 3?

4. Consider the function $f(x) = \begin{cases} x+4 & \text{if } x < 2 \\ x^2 - 1 & \text{if } 2 \leq x < 4 \\ 2 & \text{if } x \geq 4 \end{cases}$.

a) Graph $f(x)$.

b) Determine each of the following limits.

i. $\lim_{x \rightarrow 0} f(x)$

ii. $\lim_{x \rightarrow 6} f(x)$

iii. $\lim_{x \rightarrow 2.5} f(x)$

iv. $\lim_{x \rightarrow 2^-} f(x)$

v. $\lim_{x \rightarrow 2^+} f(x)$

vi. $\lim_{x \rightarrow 2} f(x)$

vii. $\lim_{x \rightarrow 4^-} f(x)$

viii. $\lim_{x \rightarrow 4^+} f(x)$

ix. $\lim_{x \rightarrow 4} f(x)$